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Red Dye No. 3: What is it, and why is it banned in makeup but OK in food?

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WASHINGTON - Color additives give the red tint to fruit punch or the green hue in mint-flavored toothpaste. They are found in thousands of consumer goods – from breakfast cereals and candies to contact lenses and cough syrup – and are often used to make the products more attractive or appetizing.

Earlier this month, a popular Easter candy was called out by nonprofit Consumer Reports for containing one such color additive, known as Red Dye No. 3, because high doses of it have been found to cause cancer in animals.

Red Dye No. 3, officially known as erythrosine or FD&C Red No. 3, is a synthetic dye derived mainly from petroleum.

In the U.S., Red 3 has been banned for decades in makeup and topical medicines after being linked to cancer. But it remains approved for use in foods, dietary supplements, and oral drugs – found in popular candies, fruit cocktails and plenty of other foods on grocery store shelves.

Consumer health groups have been calling on the U.S. Food and Drug Administration to ban its use in food for years, citing the risk of cancer and other health problems. This includes Thomas Galligan, a PhD with the Center for Science in the Public Interest (CSPI), a nonprofit group that advocates for safer and healthier foods.

Galligan said the FDA should have banned Red Dye No. 3 in foods, dietary supplements, and oral drugs "more than 30 years ago."

"FDA itself concluded in 1990 that Red 3 is an animal carcinogen, and federal laws explicitly prohibit use of color additives that cause cancer in humans or animals," Galligan told FOX Television Stations. "As a result of that decision, FDA banned use of Red 3 in cosmetics and topical drugs in 1990, but determined it would need to take separate action to ban it in foods, supplements, and oral drugs."

"The agency promised to take steps to do so, but never did," he added.

So what is the status of Red Dye No. 3? Here is what's known about the synthetic dye and regulation in the U.S.:

A history of Red Dye No. 3 in America

The U.S. has allowed dyes like Red 3 in food since 1907.

Decades later, researchers found that lab rats exposed to high doses of Red Dye No. 3 over a long period of time developed thyroid cancer. In 1990, the FDA banned the use of Red Dye No. 3 in cosmetic products like lipsticks, powders, blushes and skin care lotions, based in part off of that research.

At the time, the agency said the risk of getting cancer from Red Dye No. 3 was "no larger than 1 in 100,000," as reported by the New York Times. The FDA also told the newspaper in 1990 that it was "in the process of extending the ban" to cover foods, ingested drugs, and supplements, too.

But the CSPI and other advocates stress that those steps were never taken.

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— Thomas Galligan, PhD, with the Center for Science in the Public Interest

To date, it remains OK to use in foods and pharmaceuticals, including things like baked goods, breakfast cereals, candies, and medicines. In fact, more than 2,900 food products alone have it listed as an ingredient, according to the Environmental Working Group's food score database.

"(The FDA's) separate action never came, so for 33 years American consumers have been unnecessarily exposed to this cancer-causing color additive," Galligan said.

Last October, advocates with CSPI sent a petition to the FDA urging the agency to finally ban Red Dye No. 3 from being used in foods, dietary supplements, and oral medicines.

The petition was signed by Consumer Reports, the Environmental Working Group, Healthy Babies Bright Futures, the Center for Environmental Health, the Children's Advocacy Institute, and several other advocacy groups.

In a statement to FOX Television Stations, the FDA said it is "actively reviewing" the CSPI's petition filed for Red Dye No. 3. The public comment period recently closed on April 18.

Red Dye No. 3 and hyperactivity in some children?

In addition to a possible cancer risk posed by Red Dye No. 3, concerns have mounted about the adverse impacts on children's behavior.

A 2021 report from the California Environmental Protection Agency linked the chemical to hyperactivity and other neurobiological behaviors in some children.

Researchers placed children on a dye-free diet for several weeks and measured their behavior. The children were then given food or drinks with dyes added, including Red Dye No. 3, and measures of their behavior were recorded by a number of standardized methods, according to the report.

"These studies demonstrated clearly that some children are likely to be more adversely affected by synthetic food dyes than others," the California Environmental Protection Agency report said.

It added that animal studies indicated synthetic food dyes affect activity, memory and learning, causing changes in the neurotransmitters in the brain, and causing "microscopic changes in brain structure."

In response to the California study, the International Association of Color Manufacturers argued in a 2021 statement that the research was "based on insufficient scientific evidence."

Consumer health advocates have urged all ages to try to avoid Red 3, but Galligan said they are "particularly concerned about the impact of Red 3 on children," noting that this age group has the highest exposure to the ingredient on a bodyweight basis of all consumers.

"This is deeply worrying considering the effects Red 3 has on neurobehavior in some children and considering that childhood is a period of potentially increased susceptibility to carcinogens," Galligan said.

On its website, the FDA states that it is continuing to examine the effects of color additives on children's behavior.

"The totality of scientific evidence indicates that most children have no adverse effects when consuming foods containing color additives, but some evidence suggests that certain children may be sensitive to them," the agency states online.

In an email, an FDA official further explained the evaluation and regulation process for ingredients added to food.

"Specifically, food additives and color additives require pre-market review and approval by the FDA before they can be added to food," the agency official said in a statement. "To obtain this authorization, a manufacturer is required to supply the FDA with evidence that establishes each food or color additive is safe at its intended level of use."

If data suggests that the use of an ingredient is unsafe, the FDA moves to take certain steps to protect public health – such as revoking authorizations or approvals for certain uses, working with the industry to implement phase-out agreements and recalls, and issuing certain alerts for manufacturers and consumers.

"The FDA will continue to engage in the scientific and regulatory review of color additives and act when necessary to ensure that the products marketed to consumers are safe and properly labeled," an FDA official told FOX Television Stations.

In the meantime, the FDA official added: "Consumers who wish to limit the amount of color additives in their diets may check the food ingredient list on labels."

States move to ban Red Dye No. 3, other food additives

Last month, California lawmakers introduced a first-of-its-kind bill that would prohibit the manufacture, sale, or distribution of any food containing Red Dye No. 3, as well as titanium dioxide, potassium bromate, brominated vegetable oil, or propyl paraben – other chemicals often used to extend foods' shelf life and enhance flavor.

The bill prompted several news headlines declaring that California wants to ban Skittles and other sweet treats. But Assemblymember Jesse Gabriel, a Democrat who authored the bill, said he wants to ban the chemicals – not the candy.

Gabriel noted how plenty of alternative ingredients are available, adding that the chemicals are already banned in Europe due to possible health risks and that companies still find a way to sell candy there.

A similar bill was introduced in New York's State Senate last month as well.

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— FDA spokesperson

The National Confectioners Association said it "strongly opposed" the California bill, called AB 418, challenging the assertion that the chemicals posed health risks.

"The ingredients that would be banned under this proposal have all been approved by the U.S. Food and Drug Administration," the NCA said in a statement. "Food safety is the number one priority for U.S. confectionery companies, and we do not use any ingredients in our products that do not comply with the FDA's strictest safety standards."

But to some, including CSPI President Dr. Peter Lurie, the risks of ingredients like Red Dye No. 3 outweigh the reward.

"The primary purpose of food dyes is generally to make junk food look more attractive, especially to kids, or to trick their parents into thinking a food contains a healthy fruit like strawberries," Lurie said in a statement. "When the purpose is purely cosmetic, why is any level of risk acceptable?"

Produce pesticides: Report details the cleanest and dirtiest fruits, vegetables

This story was reported from Cincinnati. The Associated Press contributed.

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